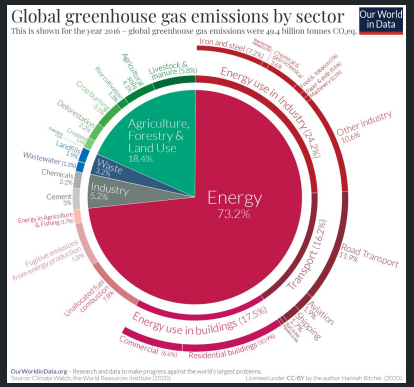

AMBIENT
SYSTEMS
Making
Through
Applied Data Science

VISION

Empower real estate leaders to achieve net zero portfolio growth through data science innovation

PROBLEM

Energy-intensive businesses face difficulty balancing emission costs with revenues and profitability







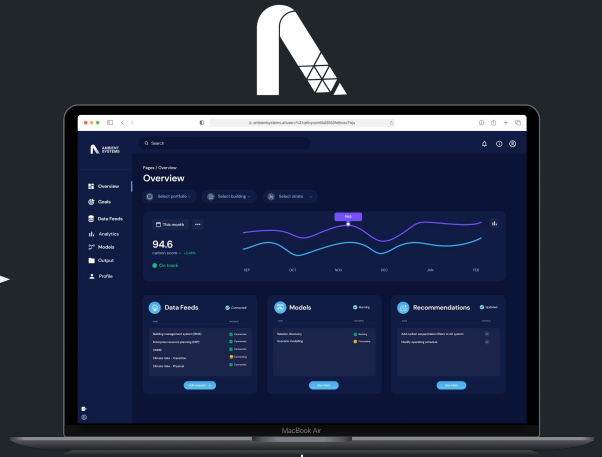
Companies can leverage data science to solve this challenge by:

01 Measuring Climate Transition Risk

02 Maximizing Revenue & Profitability

Innovative business model enabled by powerful new SaaS technology

Users



- Standardized Data Format
- Simple Pricing Plans
- Proprietary Machine Learning Models

Carbon Disclosure Organizations

Consultancies

80% Fewer Man Hours
(than conventional in-house processes or consultancies)

100% Transparent
(data and models easily verifiable by qualified auditors)

Minimized Downtime
(avoid revenue losses due to confusion, inefficiency, and dissatisfied customers)

SaaS subscription based on revenue sharing

Databases



$0 Implementation Fee
(only 3rd party vendors)

Ambient SaaS Subscription

Revenue Realized

Number of Data Feeds

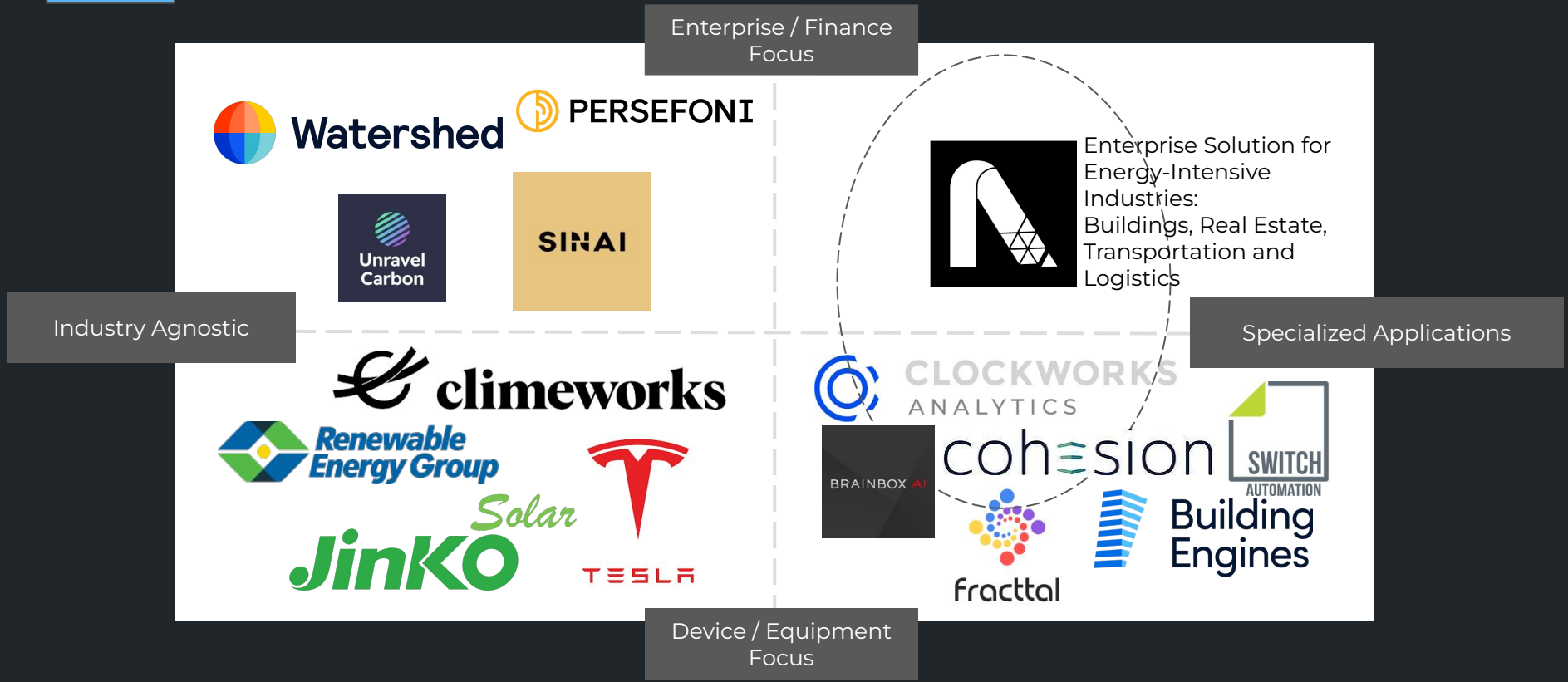
Number of Applications

ML Applications

- Process Optimization
- Predictive Maintenance
- Asset Optimization
- AI-based Demand Response
- AI-based Light Control

COMPETITION

We focus on improving financial performance of energy-intensive industries



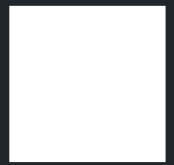
Use Case Deep Dive – Ambient Systems and Team Past Experience

Optimising for maximum Crop output and quality

Indoor Farm
Singapore, 2021

Founder sole inventor of Singapore patent for indoor farming optimization using machine learning and sensor networks

Equipment Settings



Temperature
RH
CO2
Light

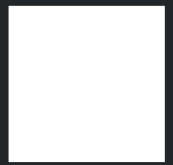
Crop Output & Quality

Optimising Energy Costs for Profitability

Gold & Nickel Mines
Africa and Central America, 2014

Saved $130MM of annual energy costs for world's biggest gold mining company using Big Data

Energy Intensity



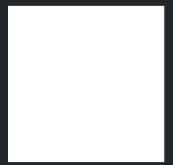
Unit Economics

Distributing Finite Energy Optimally

Sustainable Energy
Nepal, Cambodia, Indonesia, 2017

Developed 200+ solar energy, energy storage, and smart grid projects

Finite Available Power



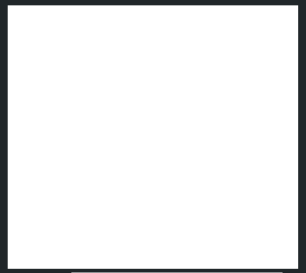
Energy Allocation per Equipment

Ambient Leadership Team

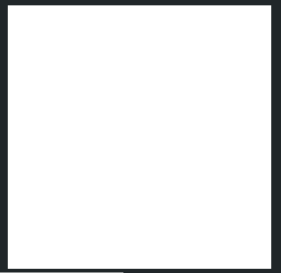


CEO
Serial entrepreneur and inventor with 5 years of experience across sustainable energy and machine learning in Canada and APAC

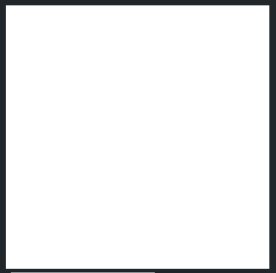
Mentors & Advisors



Dr.
Experienced technology executive, investor, and educator with 35 years of leadership experience at Oracle and Hewlett-Packard



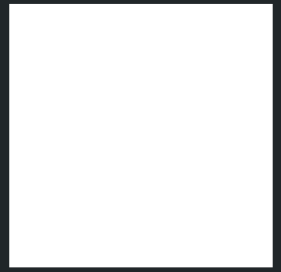
Thought leader and influencer in areas of future of work and human capital with over 50,000 followers across LinkedIn, Spotify, and Youtube



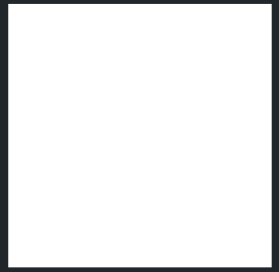
Experienced engineer and entrepreneur mentoring innovative IoT and AI spin-offs at the National University of Singapore (NUS).

ABOUT US

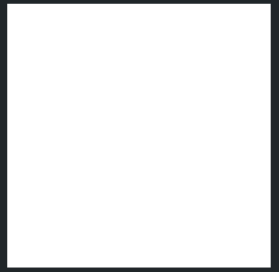
The team behind Ambient



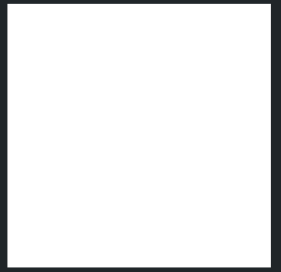
Software Engineering Intern
CS Master student from National University of Singapore School of Computing. Focused on software development.



Data Analyst Intern
Statistics Master student from National University of Singapore. Experience in applied data mining, machine learning, and deep learning.



Analyst Intern
Undergraduate majoring in Data Science and Analytics from National University of Singapore. Experience in data cleaning, visualization and machine learning.

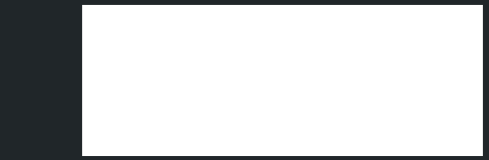


Undergraduate major in Business Administration from National University of Singapore specializing in marketing and finance. Focused on marketing campaigns.

Supported by:

Cloud Program

Google for Startups





The Ask

We will need $150k to complete a POC in Singapore in the next 12 months.

Empowering real estate leaders
to achieve net zero portfolio growth
through data science innovation

Scan to Learn More or Contact us at:

www.ambientsystems.ai

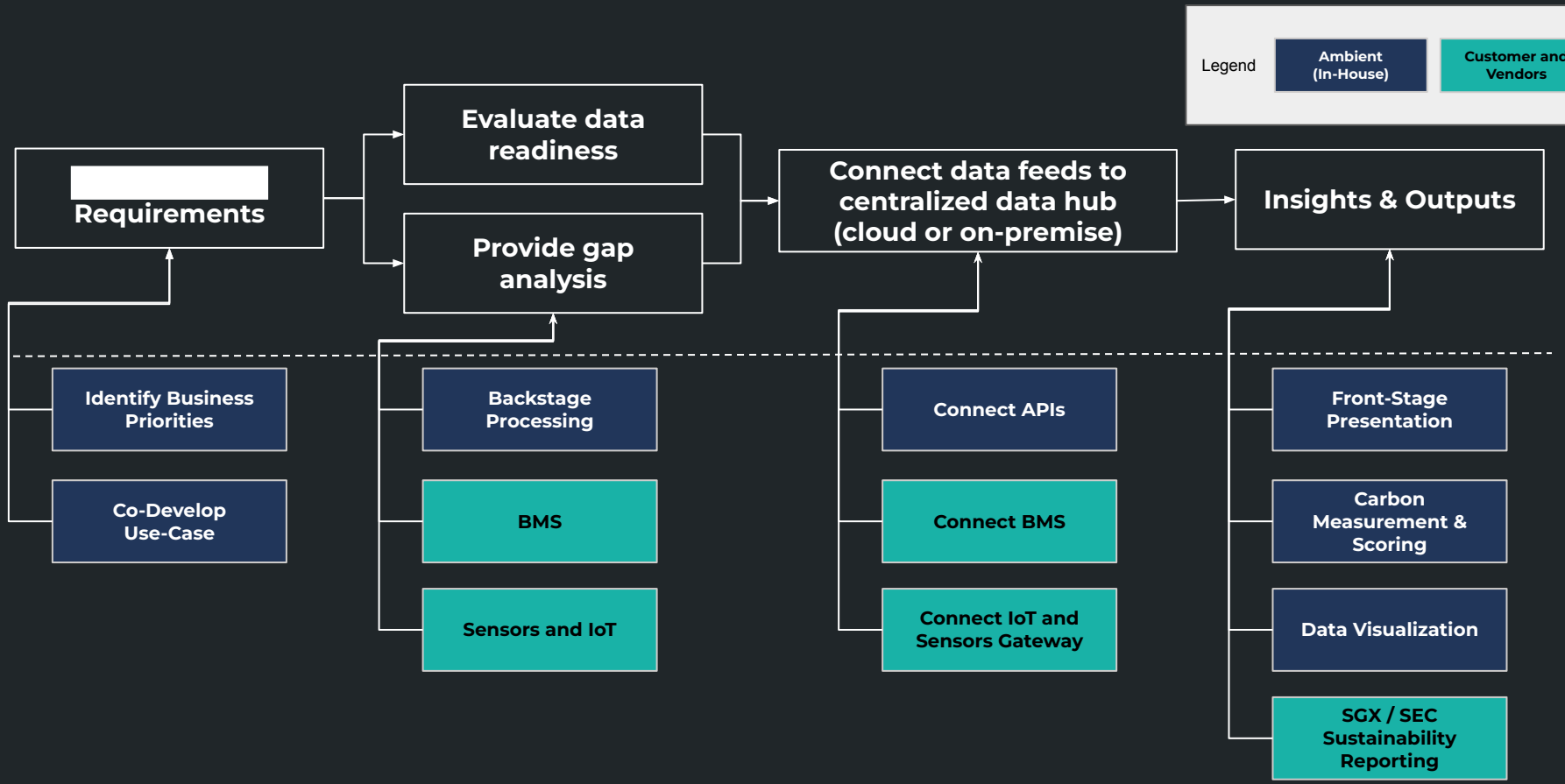
team@ambientsystems.ai

Appendix

Demo Video Link

Create a Free Demo Account at:

Quick Wins With Ambient Systems: Rapid impact with data you probably already have



PLATFORM

Data-driven Solutions for Customers

Legend: Ambient (In-House) | Customer and Vendors

Data

Technical

- BIM
- BMS
- CMMS
- BOM

Risk

- Energy Prices
- Climate Risks (Direct & Transition)
- Commodity Prices

Operational

- Electricity
- Water
- Gas
- Fuel

Data Science

Data Engineering Module
(both real-time and historical data)

[redacted]
(proprietary and patent-pending)

Output Automation Module
(includes reporting and controls)

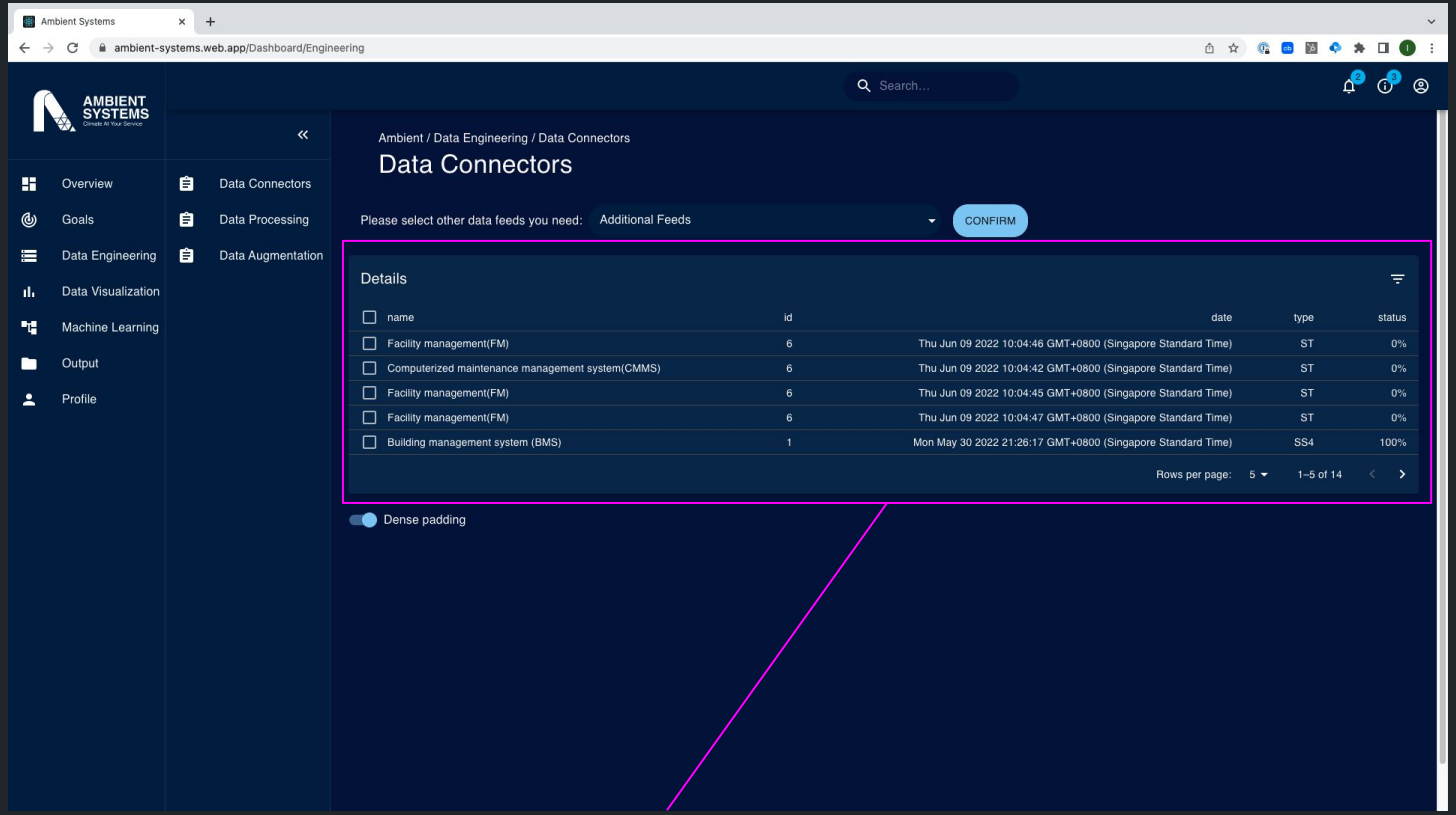
Outcomes

80% Fewer Man Hours
(than conventional in-house processes or consultancies)

100% Transparent
(data and models easily verifiable by qualified auditors)

Minimized Downtown
(avoid revenue losses due to confusion, inefficiency, and dissatisfied customers)

Step 1: Upload Historical Data or Request Real-Time Integration with Your Existing Vendor via API

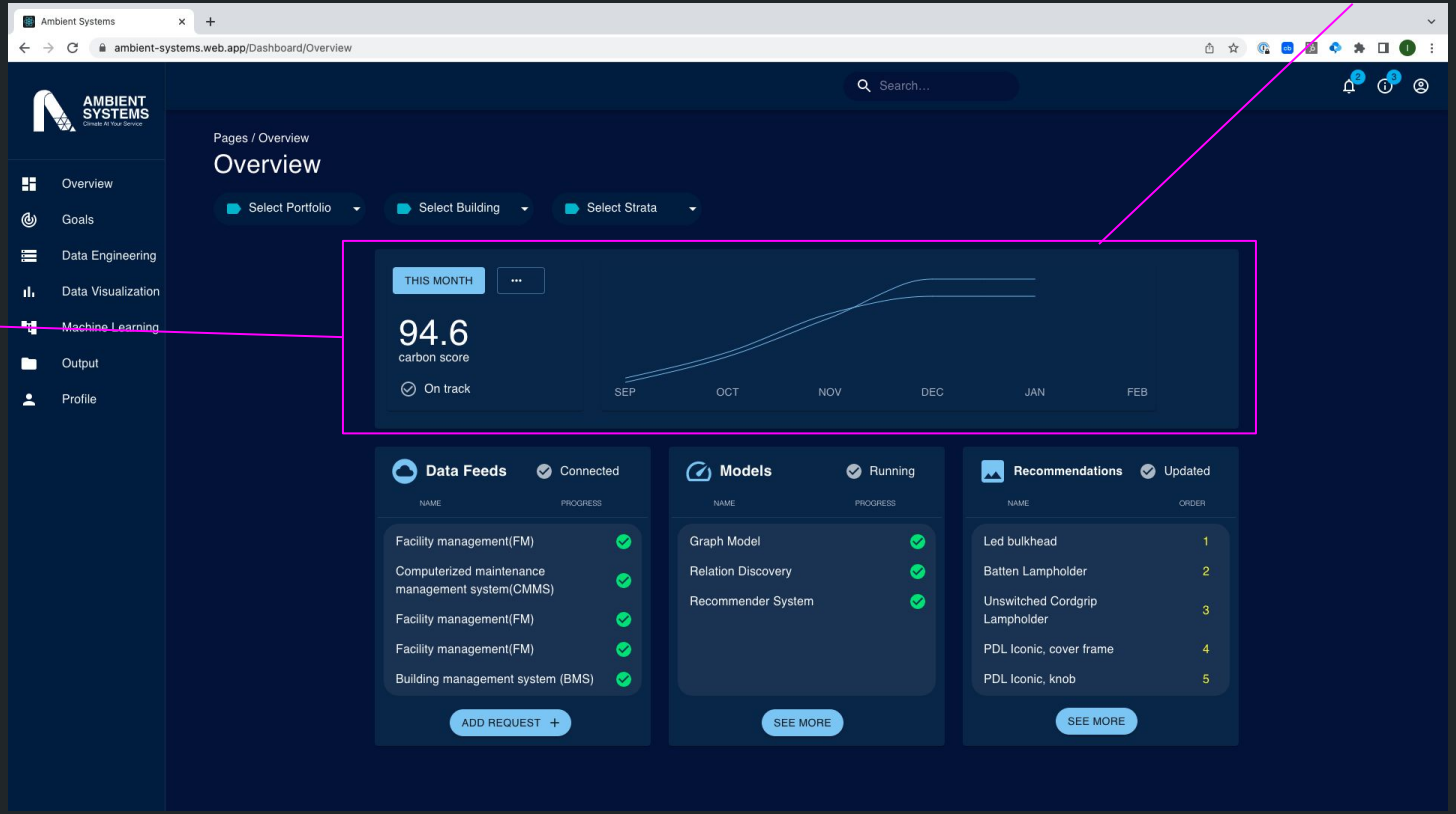


See which data feeds are connected to the platform or request new ones to be added

Step 2: See Carbon Score & Data per Location

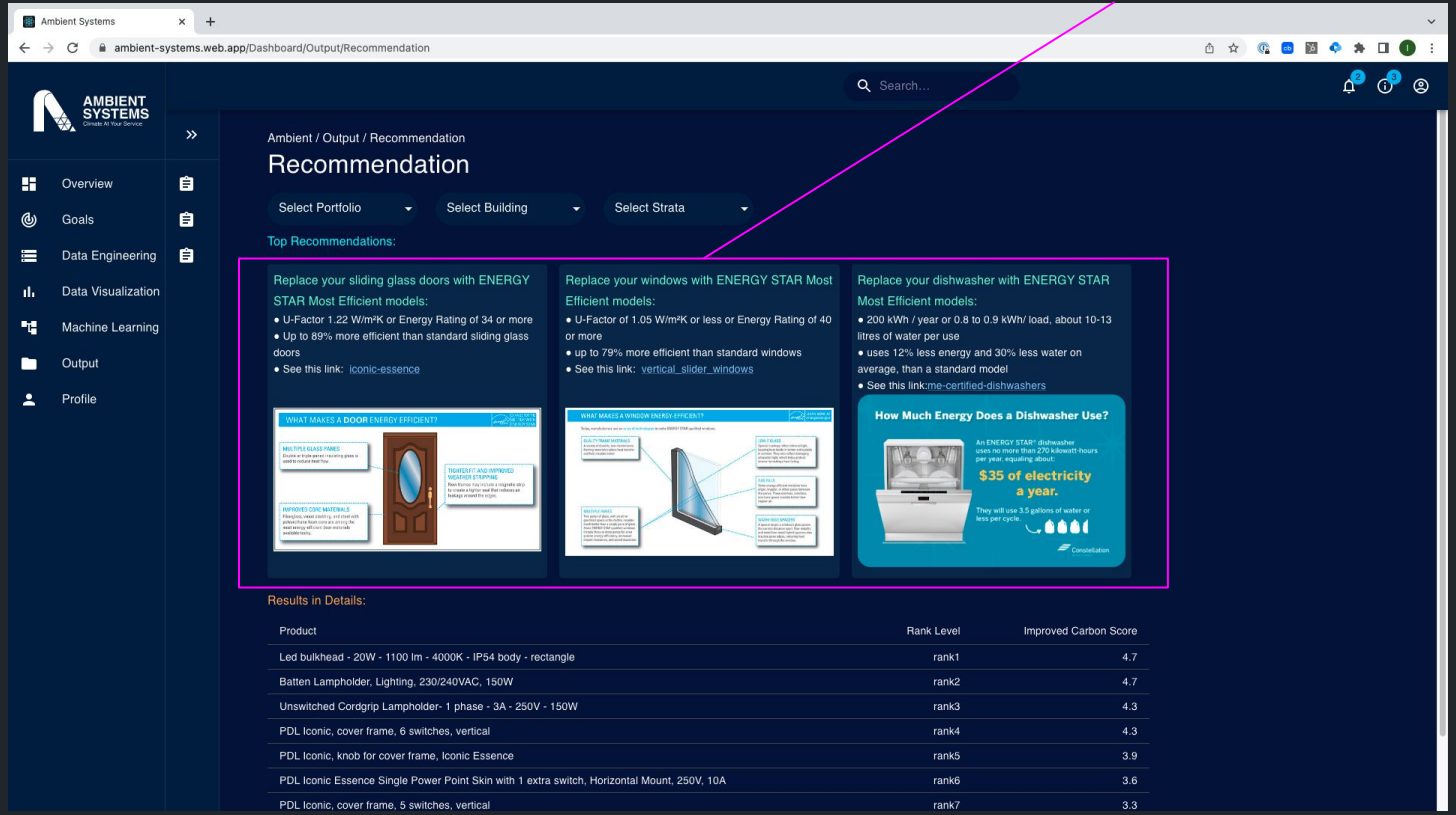
Based on one of 6 SGX-approved reporting frameworks

Carbon Score provides a quick snapshot of how well your facility is doing



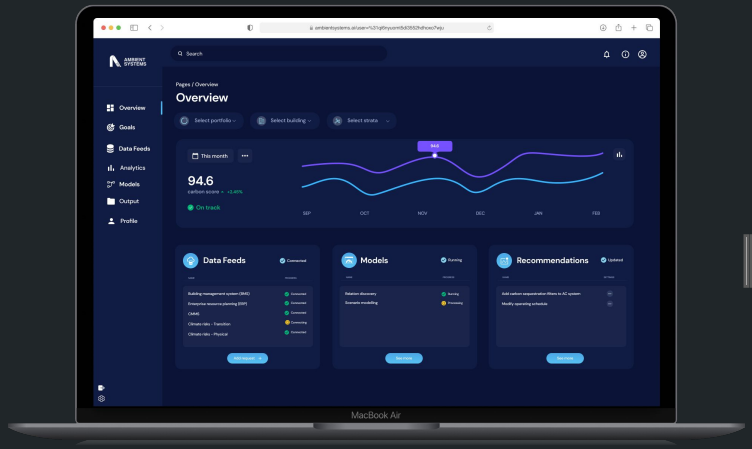
Step 3: Access Multiple Machine Learning Use-Cases

See customized recommendations on how to improve the carbon score for each your facilities



Product	Rank Level	Improved Carbon Score
Led bulkhead - 20W - 1100 lm - 4000K - IP54 body - rectangle	rank1	4.7
Batten Lampholder, Lighting, 230/240VAC, 150W	rank2	4.7
Unswitched Cordgrip Lampholder- 1 phase - 3A - 250V - 150W	rank3	4.3
PDL Iconic, cover frame, 6 switches, vertical	rank4	4.3
PDL Iconic, knob for cover frame, Iconic Essence	rank5	3.9
PDL Iconic Essence Single Power Point Skin with 1 extra switch, Horizontal Mount, 250V, 10A	rank6	3.6
PDL Iconic, cover frame, 5 switches, vertical	rank7	3.3

Automate Data Collection & Output for Sustainability Reporting



SGX-Recognized Sustainable Finance Reporting Frameworks

ESG Reporting Standards

Focused Reporting Standards

Global Reporting Initiative

TCFD TASK FORCE ON CLIMATE-RELATED FINANCIAL DISCLOSURES

SASB SUSTAINABILITY ACCOUNTING STANDARDS BOARD

CDP DISCLOSURE INSIGHT ACTION

INTEGRATED REPORTING <IR>

CDSB Climate Disclosure Standards Board

Company Roadmap and Timeline

